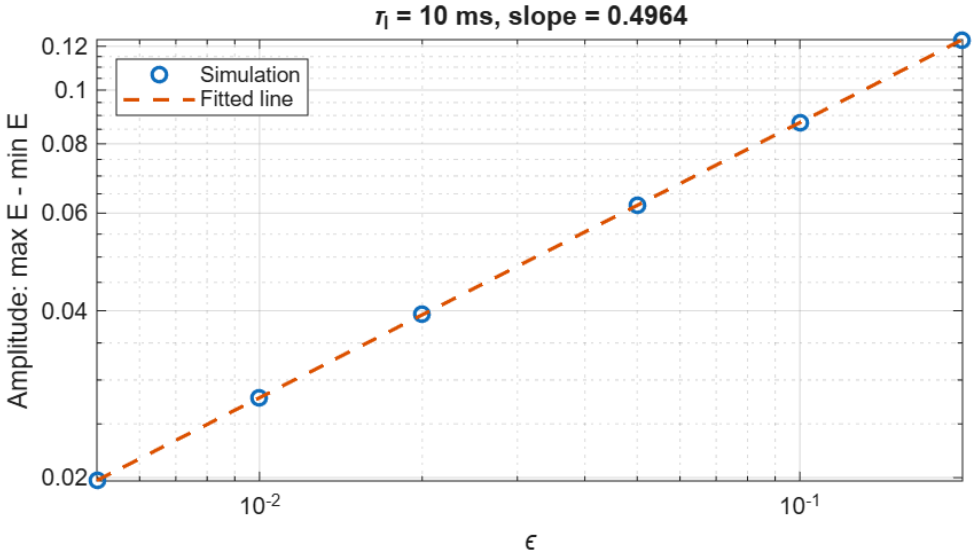
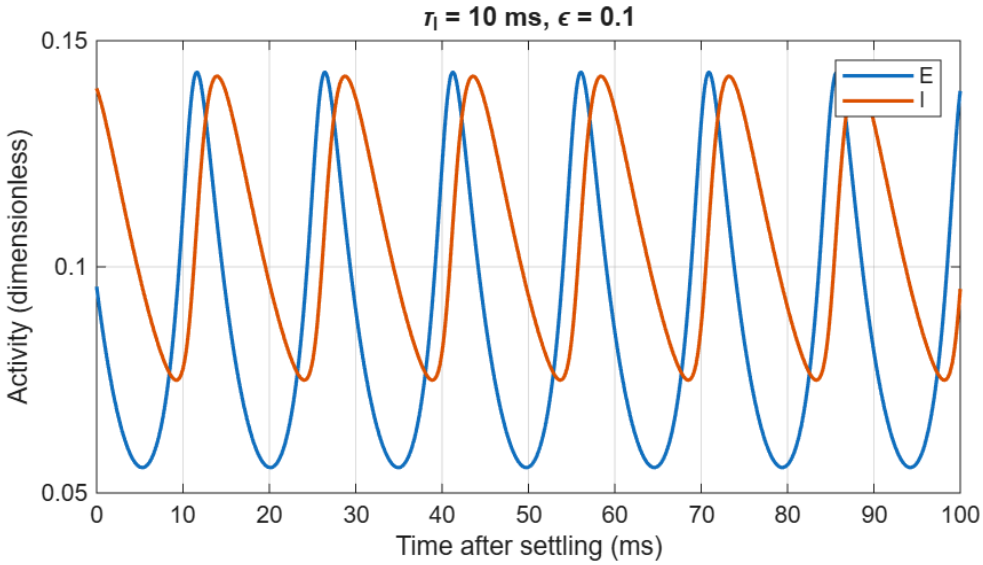
Computational Neuroscience homework has been uploaded to the following repo as per the instructions. Includes code as .m MATLAB files.

Account name: kywi4190
Repo url: https://github.com/kywi4190/CompNeuro2026
Email: kywi4190@colorado.edu

Figures

1d)





i) Results for tau_I = 10ms

epsilon	Period (ms)	Amplitude	Lag (degrees)
0.005	12.7	0.01973	73.6
0.01	12.81	0.02788	71.4
0.02	13.04	0.03939	68.4
0.05	13.71	0.06207	62.5
0.1	14.82	0.08739	56.5
0.2	17.08	0.1231	49.7

The fitted slope is (0.4964), so the amplitude grows approximately like sqrt(epsilon).

ii) Frequency comparison for epsilon = 0.02

tau_I (ms)	Simulated frequency (Hz)	Linear prediction (Hz)
2	157.1	162.8
3	133.4	138.1
5	106.4	110.1
7.5	88.02	91.13
10	76.71	79.45
15	63.02	65.30
20	54.74	56.74

Both frequencies decrease as tau_I increases. The simulated frequencies closely match the predicted frequencies.

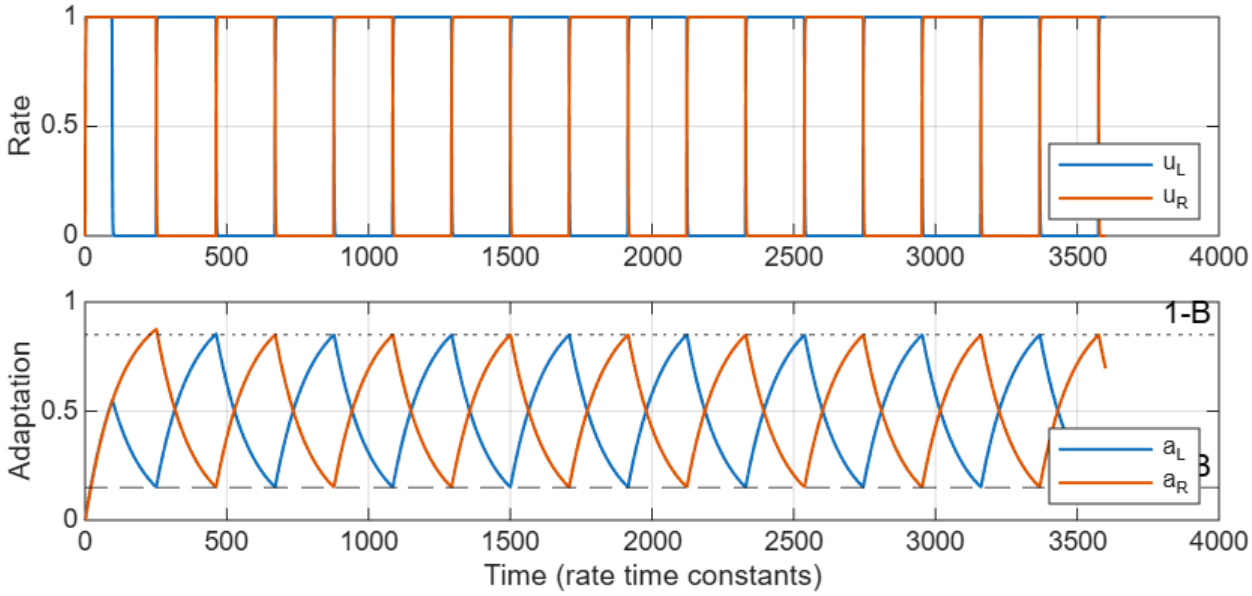
iii) Period, lag and Hopf bifurcation

As epsilon increases, the period increases and the lag in degrees decreases. Larger oscillations are more affected by the non-linear sigmoid, which changes the waveform and peak timing. The stable oscillations and

amplitude scaling close to sqrt(epsilon) support a supercritical Hopf bifurcation.

2d)

I_L=I_R=1.3, w=0.8, β=1, g=2
τ_a=120, η=1000



ii)

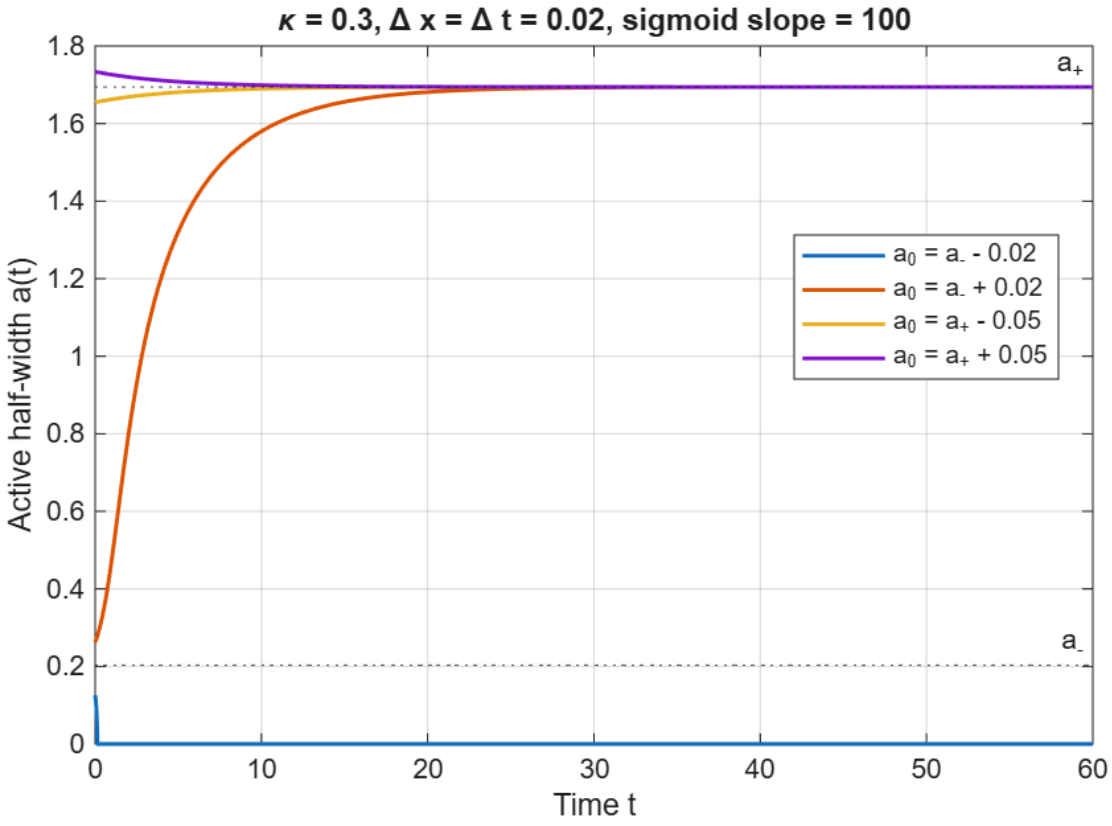
I_L	(T_L): simulated mean	(T_L): theory	(T_R): simulated mean	(T_R): theory
1.30	208	208	207	208
1.45	197	197	160	159
1.60	186	185	127	125

The simulated dominance times are very close to the theoretical predictions from part (c). As I_L increases, both dominance times decrease, but T_R decreases much more than T_L. For equal inputs, the simulated times are about 0.3% shorter than predicted. For unequal inputs, they are slightly longer, with errors below 1.3%.

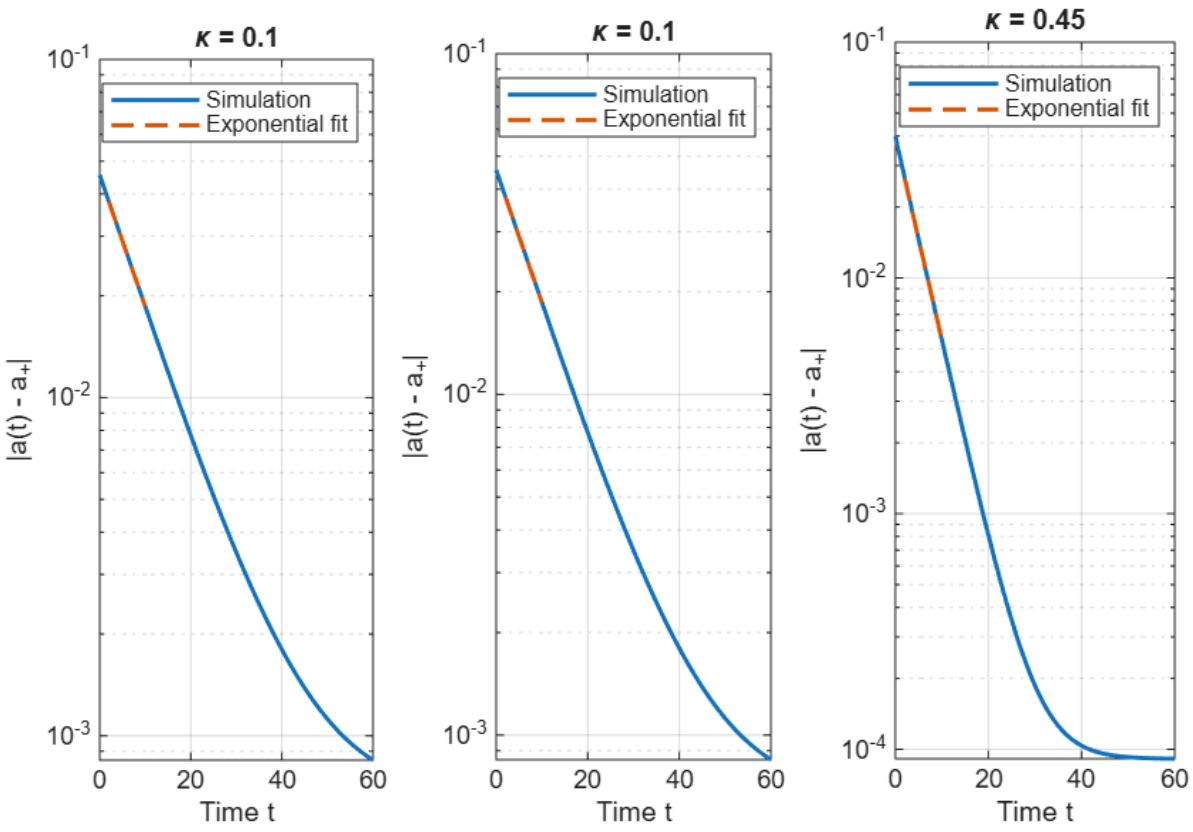
iii)

For w = 0.3, the simulated T_dom = 166, compared with the theoretical T_dom = 166. The simulation is slightly shorter by 0.0569 time units (0.0342%). The difference is very small, showing that the simulation agrees closely with the theory.

3d)



$a_0 = a_+ + 0.05$, $\Delta x = \Delta t = 0.02$, sigmoid slope = 100



i)

The narrow bump dies if started slightly too narrow, but grows toward the wide bump if started slightly too wide. Both wide bumps return to their original width.

ii)

k	**Fitted rate**	**Theory**
0.1	-0.0894	-0.09017
0.3	-0.2076	-0.2083
0.45	-0.1978	-0.1951

The fitted rates closely match theory. The small differences are likely a result of numerical approximations.

Homework 5.1

Wednesday, October 28, 2026 5:51 PM

$\eta = 4 \qquad \tau_E = 5ms \qquad w_{EI} = w_{IE} = 10 \qquad \bar{E} = \bar{I} = 0.1$

at equilibrium $\frac{dE}{dt} = \frac{dI}{dt} = 0$

$$\bar{E} = f(w_{EE}\bar{E} - w_{EI}\bar{I} + \rho_E)$$

$$\bar{I} = f(w_{IE}\bar{E} + \rho_I)$$

$$y = \frac{1}{1 + e^{-\eta x}}$$

$$\frac{1}{y} = 1 + e^{-\eta x}$$

$$e^{-\eta x} = \frac{1-y}{y}$$

$$-\eta x = \ln\left(\frac{1-y}{y}\right)$$

$$x = \frac{1}{\eta}\ln\left(\frac{y}{1-y}\right)$$

→ plug into equilibrium eqns

i) $w_{EE}\bar{E} - w_{EI}\bar{I} + \rho_E = \frac{1}{\eta}\ln\left(\frac{\bar{E}}{1-\bar{E}}\right)$

ii) $w_{IE}\bar{E} + \rho_I = \frac{1}{\eta}\ln\left(\frac{\bar{I}}{1-\bar{I}}\right)$

$$\rho_E = \frac{1}{\eta}\ln\left(\frac{\bar{E}}{1-\bar{E}}\right) - w_{EE}\bar{E} + w_{EI}\bar{I}$$

$$\rho_I = \frac{1}{\eta}\ln\left(\frac{\bar{I}}{1-\bar{I}}\right) - w_{IE}\bar{E}$$

$$\therefore p_E = 1 - \frac{\ln 9}{4} - 0.1 w_{EE}$$

$$p_I = -1 - \frac{\ln 9}{4}$$

$$f'(x) = \frac{\eta e^{-\eta x}}{(1+e^{-\eta x})^2}$$

$$f(x) = \frac{1}{1+e^{-\eta x}} \longrightarrow 1 - f(x) = \frac{e^{-\eta x}}{1+e^{-\eta x}}$$

$$f'(x) = \eta f(x) f(1 - f(x))$$

$$\bar{x}_E = w_{EE}\bar{E} - w_{EI}\bar{I} + p_E$$

$$\bar{x}_I = w_{IE}\bar{E} + p_I$$

→ equilibrium inputs to sigmoid

$$f(\bar{x}_E) = \bar{E} \qquad f(\bar{x}_I) = \bar{I}$$

$$f'(\bar{x}_E) = \eta \bar{E}(1-\bar{E})$$

$$f'(\bar{x}_I) = \eta \bar{I}(1-\bar{I})$$

$$J = \begin{pmatrix} \frac{-1 + w_{EE} f'(\bar{x}_E)}{\tau_E} & \frac{-w_{EI} f'(\bar{x}_E)}{\tau_E} \\ \frac{w_{IE} f'(\bar{x}_I)}{\tau_I} & \frac{-1}{\tau_I} \end{pmatrix}$$

$$\therefore J = \begin{pmatrix} \frac{-1 + \eta w_{EE} \bar{E}(1-\bar{E})}{\tau_E} & \frac{-\eta w_{EI} \bar{E}(1-\bar{E})}{\tau_E} \end{pmatrix}$$

$$\therefore J = \begin{pmatrix} \dfrac{-1+\eta w_{EE}\bar{E}(1-\bar{E})}{\tau_E} & \\ \eta \dfrac{w_{IE}\bar{I}(1-\bar{I})}{\tau_I} & \dfrac{-1}{\tau_I} \end{pmatrix}$$

$$\text{tr}\, J = \frac{-1+\eta w_{EE}\bar{E}(1-\bar{E})}{\tau_E} - \frac{1}{\tau_I}$$

$$\frac{-1+\eta w_{EE}\bar{E}(1-\bar{E})}{\tau_E} = \frac{1}{\tau_I}$$

$$-1+\eta w_{EE}\bar{E}(1-\bar{E}) = \frac{\tau_E}{\tau_I}$$

$$\therefore \text{tr}\, J = 0 \iff \eta w_{EE}\bar{E}(1-\bar{E}) = 1 + \frac{\tau_E}{\tau_I}$$

Hold inhibitory population fixed at $\bar{I}$.

Then use small excitatory perturbation $\delta E = E - \bar{E}$:

$$\frac{d(\delta E)}{dt} = J_{EE}\,\delta E$$

at $\text{tr}\, J = 0$ point:

$$J_{EE} = \frac{-1+\eta w_{EE}\bar{E}(1-\bar{E})}{\tau_E} = \frac{-1+(1+\frac{\tau_E}{\tau_I})}{\tau_E} = \frac{1}{\tau_I} > 0$$

$$(\delta E)' = \frac{(\delta E)}{\tau_I} \longrightarrow \delta E(t) = \delta E(0)\, e^{t/\tau_I}$$

fixed. small changes in E

(or) τ_I

With inhibition fixed, small changes in E grow exponentially. The excitatory equilibrium is therefore unstable on its own.

w_{EE}, ρ_E, ρ_I at $\tau_I = 10$:

$$4 w_{EE} \left(\frac{1}{10}\right)\left(\frac{2}{10}\right) = 1 + \frac{5}{10}$$

$$\therefore w_{EE} = \frac{25}{6} \approx 4.167$$

$$\rho_E = 1 - \frac{\ln 9}{4} - \frac{1}{10}\left(\frac{25}{6}\right)$$

$$\therefore \rho_E \approx 0.03403$$

$$\rho_I = -1 - \frac{\ln 9}{4}$$

$$\therefore \rho_I \approx -1.549$$

b) $\det(\lambda I_2 - J) = 0 = (\lambda - J_{EE})(\lambda - J_{II}) - J_{EI} J_{IE}$

$$0 = \lambda^2 - (J_{EE} + J_{II})\lambda + J_{EE} J_{II} - J_{EI} J_{IE}$$

$$0 = \lambda^2 - (\mathrm{tr} J)\lambda + \det J$$

$\longrightarrow \mathrm{tr} J = 0$

$$\lambda = \pm i \sqrt{\det J}$$

$$\omega^2 = \det J$$

$J_{EE} = \frac{1}{\tau_I}$ $\quad J_{II} = \frac{-1}{\tau_I}$ → from a)

$$\det J = J_{EE} J_{II} - J_{EI} J_{IE}$$

$$= \left(\frac{1}{\tau_I}\right)\left(\frac{-1}{\tau_I}\right) - \left(\frac{-\eta w_{EI} \bar{E}(1-\bar{E})}{\tau_E}\right)\left(\frac{\eta w_{IE} \bar{I}(1-\bar{I})}{\tau_I}\right)$$

$$= \frac{-1}{\tau_I^2} + \frac{\eta^2 w_{EI} w_{IE} \bar{E}(1-\bar{E})\bar{I}(1-\bar{I})}{\tau_E \tau_I}$$

let $B = \dfrac{\eta^2 w_{EI} w_{IE} \bar{E}(1-\bar{E})\bar{I}(1-\bar{I})}{\tau_E}$

$$\therefore \omega^2 = \frac{B}{\tau_I} - \frac{1}{\tau_I^2}$$

Evaluate ω, T, f at $\tau_I = 10$:

$$B = \frac{\eta^2 (10)(10)(0.1)(0.9)(0.1)(0.9)}{5} = \frac{12.96}{5} = 2.592$$

$$\omega^2 = \frac{2.592}{10} - \frac{1}{100} = 0.2492$$

$$\therefore \omega = \sqrt{0.2492} \approx 0.4992 \ \frac{rad}{ms} \quad \text{or} \quad \approx 499.2 \ rad/s$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{0.2492}}$$

$\therefore T \approx 12.59\,ms = 0.01259\,s$

$$f = \frac{1}{T} = \frac{1}{0.01259}$$

$\therefore f \approx 79.45\,Hz$

Find threshold for τ_I:

$$\omega^2 = \frac{B}{\tau_I} - \frac{1}{\tau_I^2} > 0$$

$$\frac{B\tau_I - 1}{\tau_I^2} > 0$$

$$B\tau_I > 1$$

$$\tau_I > \frac{1}{B}$$

$\therefore$ need $\tau_I > \frac{1}{2.592} \approx 0.3858\,ms$

The oscillation threshold is 0.3858ms and oscillations require τ_I to be strictly larger.

At the threshold $\omega = 0$.

Frequency Scaling when $\tau_I \gg \frac{1}{B}$

$$\omega^2 = \frac{B}{\tau_I}\left(1 - \frac{1}{B\tau_I}\right)$$

$\tau_I \gg \frac{1}{n} \rightarrow \frac{1}{B\tau_I} \ll 1$

$\tau_I \gg \frac{1}{B} \rightarrow \frac{1}{B\tau_I} \ll 1$

$$\omega^2 \approx \frac{B}{\tau_I}$$

$$\omega \approx \sqrt{\frac{B}{\tau_I}}$$

$$\therefore f \propto \frac{1}{\sqrt{\tau_I}}$$

For large τ_I the frequency decreases in proportion to $\frac{1}{\sqrt{\tau_I}}$.

τ_I for 40 Hz:

$$\omega = 2\pi(40)\,\text{rad/s} = 80\pi\,\text{rad/s}$$

$$= 0.08\pi\,\text{rad/ms}$$

$$\omega^2 = \frac{B}{\tau_I} - \frac{1}{\tau_I^2}$$

$$\omega^2 \tau_I^2 = B\tau_I - 1$$

$$\omega^2\tau_I^2 - B\tau_I + 1 = 0$$

$$\tau_I = \frac{B \pm \sqrt{B^2 - 4\omega^2}}{2\omega^2}$$

$$\tau_I = \frac{2.592 \pm \sqrt{(2.592)^2 - 4(0.08\pi)^2}}{2(0.08\pi)^2}$$

∴ i) $\tau_I \approx 0.3895\text{ms}$

ii) $\tau_I \approx 40.65\text{ms}$

→ both above oscillation threshold

Neither really matches the stated $GABA_A$ decay time of about 10ms. The 40.65ms is about 4 times too long while 0.3895ms is much too short.

c) $$\tau_I \frac{d(\delta I)}{dt} = -\delta I + \eta w_{IE} \bar{I}(1-\bar{I})\delta E$$

let $k = \eta w_{IE} \bar{I}(1-\bar{I}) > 0$

$$\rightarrow \tau_I \frac{d(\delta I)}{dt} = -\delta I + k\delta E$$

let $\delta E(t) = Re\left(\hat{E} e^{i\omega t}\right)$

$\delta I(t) = Re\left(\hat{I} e^{i\omega t}\right)$

→ Sinusoidal perturbations using complex amplitudes

$i\omega\tau_I \hat{I} = -\hat{I} + k\hat{E}$ → Complex oscillation amplitude and phase

$(1+i\omega\tau_I)\hat{I} = k\hat{E}$ amplitudes $\hat{E}$ and $\hat{I}$ record

$$\frac{\hat{I}}{\hat{E}} = \frac{k}{1+i\omega\tau_I} = \frac{k(1-i\omega\tau_I)}{(1+i\omega\tau_I)(1-i\omega\tau_I)}$$

$$= \frac{k(1-i\omega\tau_I)}{1+(\omega\tau_I)^2}$$

$k > 0$ so phase comes from $(1-i\omega\tau_I)$

$\frac{k}{1+(\omega\tau_I)^2} > 0$ so phase comes from $(1-i\omega\tau_I)$

$Re(1-i\omega\tau_I) = 1$

$Im(1-i\omega\tau_I) = -\omega\tau_I$

→ relative phase of $-\arctan(\omega\tau_I)$

So I is delayed relative to E with positive lag: $\boxed{\phi = \arctan(\omega\tau_I)}$

$\tau_I = 10\ ms$

$\omega = \sqrt{0.2492}\ rad/ms$

$\phi = \arctan(10\sqrt{0.2492})$

$\boxed{\therefore \phi \approx 1.373\ rad \approx 78.67^\circ}$

I is closer to a quarter cycle behind E than being in phase. Its lag is 78.67° which is only 11.33° short of a quarter cycle.

Homework 5.2

Wednesday, October 28, 2026 7:07 PM

2. $w = 0.8 \quad \beta = 1 \quad g = 2 \quad \tau_a = 120$

$I_L = I_R = I = 1.3$

let $h_L = w u_L - \beta u_R - g a_L + I_L$

$h_R = w u_R - \beta u_L - g a_R + I_R$

a) $(u_L, u_R) = (1, 0)$

$\tau_a \frac{da_L}{dt} = -a_L + 1 \qquad \tau_a \frac{da_R}{dt} = -a_R$

$\therefore \frac{da_L}{dt} = \frac{1 - a_L}{\tau_a}$

$\frac{da_R}{dt} = \frac{-a_R}{\tau_a}$

Find threshold where R turns on:

$h_R = w u_R - \beta u_L - g a_R + I$

$= w(0) - \beta(1) - g a_R + I$

$= I - \beta - g a_R$

as a_R decreases this input increases.

as a_R decreases this input increases.

Switching threshold is reached when $h_R = 0$

$$I - \beta - g a_R = 0$$

$$a_R = \boxed{\frac{I-\beta}{g} = \beta}$$

$$\beta = \frac{1.3-1}{2}$$

$$\boxed{\therefore \beta = 0.15}$$

To determine if L switches off when R turns on, we need a_L.
To do this we use a property of the established periodic alternating cycle.

During each dominance interval only one population is active:

$$u_L + u_R = 1$$

The slow-adaption approximation treats the rapid switches as instant, and adaption is unchanged during the switch.

let $s(t) = a_L(t) + a_R(t)$

$$ds = (-a_L + u_L) + (-a_R + u_R)$$

$$\tau_a \frac{dS}{dt} = (-a_L + u_L) + (-a_R + u_R)$$

$$\tau_a \frac{dS}{dt} = -(a_L + a_R) + (u_L + u_R)$$

$$\tau_a \frac{dS}{dt} = 1 - S$$

$$\rightarrow \frac{d}{dt}(S-1) + \frac{1}{\tau_a}(S-1) = 0 \quad \rightarrow \text{use integrating factor } e^{t/\tau_a}$$

$$\frac{d}{dt}\left(e^{t/\tau_a}(S-1)\right) = 0$$

$$e^{t/\tau_a}(S(t)-1) = S(0) - 1$$

$$S(t) = 1 + [S(0)-1]e^{-t/\tau_a}$$

let $P > 0$ be the full cycle period : $S(P) = S(0)$

$$S(0) - 1 = [S(0)-1]e^{-P/\tau_a}$$

$$[S(0)-1]\left(1 - e^{-P/\tau_a}\right) = 0$$

$$1 - e^{-P/\tau_a} > 0 \rightarrow \begin{matrix} P > 0 \\ \tau_a > 0 \end{matrix}$$

then $S(0) = 1$

so $a_L + a_R = 1$ throughout the periodic cycle

we have $a_R = B \qquad a_L = 1 - B$

$$\boxed{a_R = 0.15 \qquad a_L = 0.85}$$

Show L switches off when R turns on:

During this rapid transition adaption remains approximately fixed at:

$$a_L = 1 - B$$

with R and L still active:

$$h_L = w - \beta - g(1 - B) + I$$

For L to switch off need:

$$w - \beta - g(1 - B) + I < 0$$

$$w - \beta - g + gB + I < 0$$

$$B = \frac{(I - \beta)}{g} \rightarrow \quad gB = I - \beta$$

$$w - \beta - g + I - \beta + I < 0$$

$$w - g - 2\beta + 2I < 0$$

∴ L switches off when R turns on iff

$$w < g + 2\beta - 2I$$

$$g + 2\beta - 2I = 2 + 2(1) - 2(1.3) = 1.4$$

$0.8 < 1.4$ ∴ verified

Show L cannot switch itself off first:

... turns on $u_R = 0$:

Before R turns on $u_R = 0$:

$$h_L = w - g a_L + I$$

while L is dominant a_L increases so h_L decreases.
Its smallest value before R turns on is at:

$$a_L = 1 - B$$

$$w - g(1 - B) + I > 0 \longrightarrow \text{so L is active till then}$$

$$w - g + gB + I > 0$$

$$w - g + I - \beta + I > 0$$

$$w - g - \beta + 2I > 0$$

$\therefore$ L cannot switch itself off first provided $w > g + \beta - 2I$

$$g + \beta - 2I = 2 + 1 - 2(1.3) = 0.4$$

$0.8 > 0.4$ $\therefore$ verified

Both conditions hold. L stays active while R's adaption decreases to $B = 0.15$.

when R turns on its inhibition makes L input negative so L switches off.

. dominant

negative so L

b) Set $t=0$ directly after L becomes dominant

$a_L(0) = B \qquad a_R(0) = a_{hi}$

$$\frac{da_L}{dt} + \frac{1}{\tau_a} a_L = \frac{1}{\tau_a}$$

$$\frac{d}{dt}\left(e^{t/\tau_a} a_L\right) = \frac{1}{\tau_a} e^{t/\tau_a}$$

$$e^{t/\tau_a} a_L(t) - B = e^{t/\tau_a} - 1$$

$$e^{t/\tau_a} a_L(t) = e^{t/\tau_a} + B - 1$$

$$\boxed{a_L(t) = 1 - (1-B)e^{-t/\tau_a}}$$

$$\frac{da_R}{dt} + \frac{1}{\tau_a} a_R = 0$$

$$\frac{d}{dt}\left(e^{t/\tau_a} a_R\right) = 0$$

$$e^{t/\tau_a} a_R(t) - a_{hi} = 0$$

$$\boxed{a_R(t) = a_{hi} e^{-t/\tau_a}}$$

at $t = T$ R hits threshold and becomes next winner.

use symmetry for:

$a_R(T) = B$

$a_L(T) = a_{hi}$ $\quad a_R(T) = B$

$$a_{hi} = 1 - (1-B)e^{-T/\tau_a}$$

$$B = a_{hi} e^{-T/\tau_a}$$

let $x = e^{-T/\tau_a}$

need $T > 0$ so $0 < x < 1$

i) $a_{hi} = 1 - (1-B)x$ $\quad$ ii) $B = a_{hi} x$

$$B = x(1-(1-B)x)$$

$$B = x - (1-B)x^2$$

$$(1-B)x^2 - x + B = 0$$

$$(1-B)x^2 - (1-B)x - Bx + B = 0$$

$$(1-B)x(x-1) - B(x-1) = 0$$

$$(x-1)[(1-B)x - B] = 0$$

$$x = 1, \ \frac{B}{1-B}$$

$$e^{-T/\tau_a} = 1 \rightarrow T = 0$$

$$\therefore x = \frac{B}{1-B}$$

$$a_{hi} = 1 - (1-B)\left(\frac{B}{1-B}\right) = 1 - B$$

$$\boxed{a_{hi} = 1 - B = 0.85}$$

$$e^{-T/\tau_a} = \frac{B}{1-B}$$

$$\frac{-T}{\tau_a} = \ln\left(\frac{B}{1-B}\right) \longrightarrow -\ln r = \ln\left(\frac{1}{r}\right)$$

$$\boxed{\therefore T_{dom} = T = \tau_a \ln\left(\frac{1-B}{B}\right)}$$

$B = 0.15 \qquad \tau_a = 120$

$$T_{dom} = 120 \ln\left(\frac{0.85}{0.15}\right)$$

$$\boxed{\therefore T_{dom} \approx 208.2}$$

c) $I_L = 1.6 \qquad I_R = 1.3$

$$B_j = \frac{I_j - \beta}{J}$$

$$\rightarrow B_L = \frac{1.6 - 1}{2} = 0.3$$

$$\rightarrow B_R = \frac{1.3 - 1}{2} = 0.15$$

1. dominance times of L and R

let T_L, T_R be dominance times of L and R

let $x_L = e^{-T_L/\tau_a}$ $\quad x_R = e^{-T_R/\tau_a}$

let a_{Lhi}, a_{Rhi} be each population's adaption at the end of its own dominance interval

right after L turns on:

$$a_L(0) = B_L \qquad a_R(0) = a_{Rhi}$$

$$a_L(t) = 1 - (1 - B_L)e^{-t/\tau_a}$$

$$a_R(t) = a_{Rhi}\, e^{-t/\tau_a}$$

→ From b)

at $t = T_L$:

$$a_{Lhi} = 1 - (1 - B_L)x_L$$

$$B_R = a_{Rhi}\, x_L$$

Reset to s=0 when R turns on:

$$a_R(0) = B_R \qquad a_L(0) = a_{Lhi}$$

$$a_R(s) = 1 - (1 - B_R)e^{-s/\tau_a}$$

$$a_L(s) = a_{Lhi}\, e^{-s/\tau_a}$$

at $s = T_R$:

$$a_{Rhi} = 1 - (1 - B_R)x_R$$

$$B_L = a_{Lhi}\, x_R$$

$a_{Rhi} = 1 - (\quad)$

$B_L = a_{Lhi}\, x_R$

$$\therefore B_L = x_R\left(1-(1-B_L)x_L\right)$$
$$B_R = x_L\left[1-(1-B_R)x_R\right]$$

$$x_R = \frac{B_L}{1-(1-B_L)x_L}$$

$$B_R = x_L\left[1-(1-B_R)\left(\frac{B_L}{1-(1-B_L)x_L}\right)\right]$$

$$B_R\left[1-(1-B_L)x_L\right] = x_L\left[1-(1-B_L)x_L - B_L(1-B_R)\right]$$

$$B_R - B_R(1-B_L)x_L = x_L - (1-B_L)x_L^2 - B_L(1-B_R)x_L$$

$$(1-B_L)x_L^2 - \left[1-B_L(1-B_R)+B_R(1-B_L)\right]x_L + B_R = 0$$

$$1-B_L(1-B_R)+B_R(1-B_L) = 1-B_L+B_LB_R+B_R-B_LB_R$$
$$= 1-B_L+B_R$$

$$(1-B_L)x_L^2 - (1-B_L+B_R)x_L + B_R = 0$$

$$(1-B_L)x_L^2 - (1-B_L)x_L - B_Rx_L + B_R = 0$$

$$(1-B_L)x_L(x_L-1) - B_R(x_L-1) = 0$$

$$(x_L-1)\left[(1-B_L)x_L - B_R\right] = 0$$

$$x_L = 1 \quad , \quad \frac{B_R}{1-B_L}$$

$$x_L = 1: \quad x_R = \frac{B_L}{1-(1-B_L)} = \frac{B_L}{B_L} = 1$$

thin $T_L = T_R = 0$

$$\therefore x_L = \frac{B_R}{1-B_L} = \frac{0.15}{0.70} \approx 0.2143$$

$$x_R = \frac{B_L}{1-(1-B_L)\frac{B_R}{1-B_L}} = \frac{B_L}{1-B_R}$$

$$\therefore x_R = \frac{B_L}{1-B_R} = \frac{0.3}{0.85} \approx 0.3529$$

$$e^{-T_L/\tau_a} = \frac{B_R}{1-B_L}$$

$$-\frac{T_L}{\tau_a} = \ln\left(\frac{B_R}{1-B_L}\right)$$

$$\therefore T_L = \tau_a \ln\left(\frac{1-B_L}{B_R}\right)$$

$$e^{-T_R/\tau_a} = \frac{B_L}{1-B_R}$$

$$-\frac{T_R}{\tau_a} = \ln\left(\frac{B_L}{1-B_R}\right)$$

$$\therefore T_R = \tau_a \ln\left(\frac{1-B_K}{B_L}\right)$$

$$T_L = 120 \ln\left(\frac{1-0.3}{0.15}\right)$$

$$\therefore T_L \approx 184.9$$

$$T_R = 120 \ln\left(\frac{1-0.15}{0.30}\right)$$

$$\therefore T_R \approx 125.0$$

Fraction L is dominant:

$$F_L = \frac{T_L}{T_L + T_R} = \frac{184.9}{184.9 + 125}$$

$$F_L \approx 0.5966 = 59.66\%$$

Left percept dominates around 60% of the time.

$$T_{dom} \approx 208.15$$

$$T_{dom} - T_L \approx 208.15 - 184.85 = 23.3$$

$$T_{dom} - T_R \approx 208.15 - 124.97 = 83.18$$

R dominance shrinks more (83.18 units) compared (23.3 units).

R dominance shrinks more to L (23.3 units).

Level's second proposition predicts that strengthening one eye leaves its own dominance duration unchanged, while the other eye's dominance reduces in duration. Here, strengthening L mainly decreases R dominance, though also slightly decreases its own. It somewhat agrees with this proposition.

Homework 5.3

Thursday, October 29, 2026 4:18 PM

3. $\frac{\partial u}{\partial t} = -u + \int_{-\infty}^{\infty} w(x-y) H(u(y,t)-k)\,dy$ $\qquad w(x) = 2e^{-|x|} - e^{-|x|/2}$

a) $\boxed{w(0) = 2-1 = 1}$

Set $x \geq 0$ so $|x| = x$:

$w(x) = 2e^{-x} - e^{-x/2} = \underbrace{e^{-x/2}}_{\text{positive}}(2e^{-x/2} - 1)$

need: $2e^{-x/2} - 1 = 0 \longrightarrow e^{-x/2} = \frac{1}{2}$

$-\frac{x}{2} = -\ln 2$

$x = 2\ln 2$

$e^{-x/2}$ decreases as x increases so:

$w(x) > 0$ for $0 \leq x < 2\ln 2$

$w(x) < 0$ for $x > 2\ln 2$

$\int_{-\infty}^{\infty} w(x)\,dx = 2\int_0^{\infty} (2e^{-x} - e^{-x/2})\,dx$ $\longrightarrow$ using evenness

$= 2\left[-2e^{-x} + 2e^{-x/2}\right]_0^{\infty}$

$= 2[0 - (-2+2)]$

$\therefore \boxed{\int_{-\infty}^{\infty} w(x)\,dx = 0}$

Find $W(x)$ and its max:

for $x \geq 0$:

$$W(x) = \int_0^x w(z)\,dz$$

$$= \int_0^x \left(2e^{-z} - e^{-z/2}\right) dz$$

$$= \left[-2e^{-z} + 2e^{-z/2}\right]_0^x$$

$$= -2e^{-x} + 2e^{-x/2}$$

$$W(x) = 2\left(e^{-x/2} - e^{-x}\right) \qquad x \geq 0$$

for $x < 0$: $\quad w(z) = 2e^{z} - e^{z/2}$

$$W(x) = \int_0^x \left(2e^{z} - e^{z/2}\right) dz$$

$$= \left[2e^{z} - 2e^{z/2}\right]_0^x$$

$$= 2e^{x} - 2e^{x/2}$$

$$\therefore W(x) = \begin{cases} 2\left(e^{-x/2} - e^{-x}\right) & x \geq 0 \\ 2\left(e^{x} - e^{x/2}\right) & x < 0 \end{cases}$$

note: $W(-x) = -W(x)$

use $W'(x) = w(x)$

$w > 0$ before $2\ln 2$ and negative

use $W(x) = w(x)$

on positive axis $w > 0$ before $2\ln 2$ and negative afterwards. So W increases up to $2\ln 2$ then decreases.

So max is: $W(2\ln 2) = 2\left(e^{-\ln 2} - e^{-2\ln 2}\right)$

$$= 2\left(\frac{1}{2} - \frac{1}{4}\right)$$

$$= \frac{1}{2}$$

note $W(x) < 0$ for $x < 0$

$$\therefore \max\left(W(x)\right) = \frac{1}{2} \quad \text{at} \quad x = 2\ln 2$$

For stationary bump $u(x,t) = U(x)$ so $u_t = 0$. The Heaviside Function equals one inside active interval and zero outside.

$$U(x) = \int_{-a}^{a} w(x-y)\,dy$$

Set $z = x - y$

$dy = -dz$

$$U(x) = \int_{x+a}^{x-a} w(z)(-dz)$$

$$= \int_{x-a}^{x+a} w(z)\,dz$$

$$U(x) = W(x+a) - W(x-a) \longrightarrow W \text{ odd}$$

$$U(-x) = W(a-x) - W(-x-a)$$

$$= -W(x-a) + W(x+a)$$

$$= U(x)$$ → bump is even

continuous profile crosses the threshold at $x=a$:

$$U(a) = k$$

$$U(a) = W(2a) - W(0) = W(2a)$$

→ $2a > 0$

$$\therefore k = W(2a) = 2(e^{-a} - e^{-2a})$$

Solve for half-widths

edge condition is a quadratic in e^{-a}

let $q = e^{-a}$

Finite positive half-width corresponds to $0 < q < 1$.

Edge condition becomes:

$$k = 2q - 2q^2$$

$$2q^2 - 2q + k = 0$$

$$q = \frac{2 \pm \sqrt{(-2)^2 - 4(2)k}}{4}$$

$$= \frac{2 \pm \sqrt{4 - 8k}}{4}$$

$$= \frac{1 \pm \sqrt{1 - 2k}}{2}$$

$$\boxed{e^{-a} = \frac{1}{2}\left(1 \pm \sqrt{1 - 2k}\right)}$$

let $S = \sqrt{1 - 2k}$

for $0 < k < \frac{1}{2}$ we have $0 < S < 1$.

Since $a = -\ln q$ decreases as q increases, the larger root gives the smaller width.

$$\boxed{a_- = -\ln\left(\frac{1+S}{2}\right) = \ln\left(\frac{2}{1+S}\right) \longrightarrow \text{narrow bump}}$$

$$\boxed{a_+ = -\ln\left(\frac{1-S}{2}\right) = \ln\left(\frac{2}{1-S}\right) \hookrightarrow \text{wide bump}}$$

check that profiles have the assumed active interval

for $0 \leq x \leq a$ oddness of W means:

$$U(x) = W(a+x) + W(a-x)$$

let $p = e^{-(a+x)/2}$ $\quad r = e^{-(a-x)/2}$ $\quad q = e^{-a} = pr$

let $p = e^{-(a+x)/2}$ $\quad r = e^{-(a-x)/2}$ $\quad q = e^{-a} = pr$

let $s = p + r$

$$u(x) = 2(p + r - p^2 - r^2)$$

$$u(x) - k = 2(s - (s^2 - 2q) - (q - q^2)) \longrightarrow k = 2(q - q^2)$$

$$= 2(s - s^2 + q + q^2)$$

$$= 2(1 + q - s)(s + q)$$

$$= 2(1 + pr - p - r)(p + r + q)$$

$$u(x) - k = 2(1-p)(1-r)(p + r + q)$$

Here $0 < p$ and $r \leq 1$ so:

$$u(x) \geq k \quad \text{for} \quad 0 \leq x < a$$

for $x > a$:

let $v = e^{-(x-a)/2}$ $\quad 0 < v < 1$

$$u(x) = w(x+a) - w(x-a)$$

$$= 2(qv - q^2v^2) - 2(v - v^2)$$

$$= 2(1-q)[(1+q)v^2 - v)$$

Subtract $k = 2q(1-q)$

$$u(x) - k = 2(1-q)[(1+q)v^2 - v - q]$$

$$= 2(1-q)(v-1)[(1+q)v + q)$$

$$< 0$$

... results on the negative side

< 0

evenness gives same results on the negative side

So Finite non-zero bumps exist for $0 < k \leq \frac{1}{2}$.

Calculate widths and peak heights at $k = 0.3$

$$S = \sqrt{1 - 2(0.3)} = \sqrt{0.4}$$

$$e^{-a} = \frac{1 + \sqrt{0.4}}{2}$$

$$a_- = -\ln\left(\frac{1 + \sqrt{0.4}}{2}\right)$$

$$\therefore a_- \approx 0.2031$$

$$a_+ = -\ln\left(\frac{1 - \sqrt{0.4}}{2}\right)$$

$$\therefore a_+ \approx 1.694$$

widths

$$U(0) = W(a) - W(-a) = 2W(a)$$

$$U(0) = 4\left(e^{-a/2} - e^{-a}\right)$$

$$= 4\left(\sqrt{q} - q\right)$$

$$U_-(0) = 4\left(\sqrt{\frac{1 + \sqrt{0.4}}{2}} - \frac{1 + \sqrt{0.4}}{2}\right)$$

$$\therefore U_-(0) \approx 0.3489$$

$$U_+(0) = 4\left(\sqrt{\frac{1 - \sqrt{0.4}}{2}} - \frac{1 - \sqrt{0.4}}{2}\right)$$

peak heights

$u_f(0) = \eta\left(1 - \frac{\quad}{2}\right) \quad 2\ 1$

$\therefore u_f(0) \approx 0.9797$

b) Since $a > 0$:

$$w(2a) = 2e^{-2a} - e^{-a}$$

$$k = 2e^{-a} - 2e^{-2a} \rightarrow \text{from a)}$$

$$2e^{-2a} = 2e^{-a} - k$$

$$w(2a) = (2e^{-a} - k) - e^{-a}$$

$$= e^{-a} - k$$

$$\therefore w(2a_+) = e^{-a_+} - k$$

we have: $\lambda_e(a) = \frac{2w(2a)}{w(0) - w(2a)} \rightarrow w(0) = 1$

$$\lambda_e(a) = \frac{2w(2a)}{1 - w(2a)}$$

From before we have $s = \sqrt{1-2k}$

$$s^2 = 1 - 2k \rightarrow k = \frac{1-s^2}{2}$$

$$e^{-a_-} = \frac{1+s}{2}$$

$$\rightarrow w(2a_-) = e^{-a_-} - k$$

$$\longrightarrow w(2n_-) = e^{-n_-} - k$$

$$= \frac{1+s}{2} - \frac{1-s^2}{2}$$

$$= \frac{s+s^2}{2}$$

$$= \frac{s(1+s)}{2}$$

$$\lambda_c(n_-) = \frac{2\left[s(1+s)\left(\frac{1}{2}\right)\right]}{1-s(1+s)\left(\frac{1}{2}\right)}$$

$$\therefore \lambda_c(n_-) = \frac{s(1+s)}{1-s(1+s)/2}$$

$$e^{-n_+} = \frac{1-s}{2}$$

$$w(2n_+) = e^{-n_+} - k$$

$$= \frac{1-s}{2} - \frac{1-s^2}{2}$$

$$= \frac{-s+s^2}{2}$$

$$= \frac{-s(1-s)}{2}$$

$$\lambda_c(n_+) = \frac{2[-s(1-s)/2]}{1+s(1-s)/2}$$

$$\therefore \lambda_e(a_+) = \frac{-S(1-S)}{1+S(1-S)/2}$$

at $k=0.3$:

$$w(2a_-) = \frac{1+\sqrt{0.4}}{2} - 0.3 \approx 0.5162$$

$$w(2a_+) = \frac{1-\sqrt{0.4}}{2} - 0.3 \approx -0.1162$$

$$\rightarrow \lambda_e(a_-) = \frac{2(0.5162)}{1-0.5162}$$

$$\therefore \lambda_e(a_-) \approx 2.134$$

$$\rightarrow \lambda_e(a_+) = \frac{2(-0.1162)}{1+0.1162}$$

$$\therefore \lambda_e(a_+) \approx -0.2083$$

An even perturbation changes width symmetrically and its amplitude changes like $e^{\lambda_e t}$.

So: $\lambda_e > 0 \rightarrow$ growth

$\lambda_e < 0 \rightarrow$ decay

$\therefore$ narrow bump is unstable $(\lambda_e(a_-) > 0)$

wide bump is stable $(\lambda_e(a_+) < 0)$

odd-mode eigenvalue

An odd perturbation shifts the bump's position.

Shifting a stationary bump produces another stationary bump.

for small displacement ε:

$$U(x-\varepsilon) = U(x) - \varepsilon U'(x) + O(\varepsilon^2) \longrightarrow \text{Taylor Expansion}$$

$$\delta u(x) = -\varepsilon U'(x) \longrightarrow \text{perturbation from small shift}$$

U is even so U' is odd.

Shifted profile remains stationary so:

$$e^{\lambda_0 t} = 1$$

$$\boxed{\lambda_0 = 0}$$

The odd mode translates bump without changing width. Its eigenvalue is zero because the model has no preferred position.

limit $k \to \frac{1}{2}$

$$\lim_{k \to \frac{1}{2}^-} s = \sqrt{1-2k} \to 0$$

$$a_- = \ln\left(\frac{2}{1+s}\right) \to \ln 2$$

$$a_+ = \ln\left(\frac{2}{1-s}\right) \to \ln 2$$

$a_-, a_+ \to \ln 2 \approx 0.6931$

The bumps merge at $a = \ln 2$.

$$\lambda_e(a_+) = \frac{-s(1-s)}{1+s(1-s)/2}$$

$$\lambda_e(a_+) \to 0^-$$

$$\frac{\lambda_e(a_+)}{-s} = \frac{1-s}{1+s(1-s)/2} \to 1$$

$$\therefore \lambda_e(a_+) \sim -\sqrt{1-2k}$$
as $k \to \frac{1}{2}^-$

The wide bumps even eigenvalue approaches zero from below like $-\sqrt{1-2k}$ so width disturbances decay increasingly slowly.

c) Why following edges is sufficient

For a perturbation that preserves a connected active interval, let its edges be $x_L(t)$ and $x_R(t)$.

The Heaviside output is:

$$H(u(y,t)-k) = \begin{cases} 1 & x_L(t) < y < x_R(t) \\ 0 & \text{outside} \end{cases}$$

So the recurrent input is:

So the recurrent input is:

$$\int_{x_L(t)}^{x_R(t)} w(x-y)\,dy$$

This input depends on the two edges, not how far lies above threshold in the interval.

If a small perturbation doesn't change the active interval then it contributes nothing to the integral.

Subtracting the two field equations give:

$$\frac{\partial \delta u}{\partial t} = -\delta u$$

$$\longrightarrow \delta u(x,t) = \delta u(x,0)\, e^{-t}$$

The two edges determine which neurons are active and so determine the recurrent input. Perturbations that don't move the edges decay through the $-u$ term.

Perturbed narrow bump

We already know from its eigenvalue that a narrow bump moves away from its equilibrium width. To determine where it goes we use the threshold condition.

For symmetric bump with half-width $a(t)$:

$$u(a(t), t) = k$$

$$u(a(t), t) = k$$

$$u_t(a(t), t) + u_x(a(t), t)\frac{da}{dt} = 0$$

$$u_t(a(t), t) = -k + \int_{-a(t)}^{a(t)} w(a(t) - y)\, dy \quad \longrightarrow \text{from field equation}$$

$$= -k + W(2a(t))$$

$$W(2a) - k + u_x(a, t)\frac{da}{dt} = 0$$

$$\boxed{\frac{da}{dt} = \frac{W(2a) - k}{-u_x(a, t)}}$$

At a regular right edge the profile crosses downward through threshold:

$$u_x(a, t) < 0$$

then $\operatorname{Sign}\left(\frac{da}{dt}\right) = \operatorname{Sign}(W(2a) - k)$

From part a) $W(2a)$ starts at zero, rises to max then decreases toward zero. $W(2a) = k$ at a_- and a_+.

Half-width	$W(2a - k)$	motion of edges
$0 < a < a_-$	negative	bump shrinks
$a_- < a < a_+$	positive	bump expands
$a > a_+$	negative	bump shrinks

$\therefore$ Slightly too narrow: the bump shrinks and disappears.

$\therefore$ Slightly too narrow: the bump shrinks and disappears.

Slightly too wide: the bump expands until it reaches the stable wide bump with half-width a_+.